Decentralized Finance

Synthetic And Derivatives; Portfolio Management; Insurance; Information and Data Markets

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Wallet Account Overview

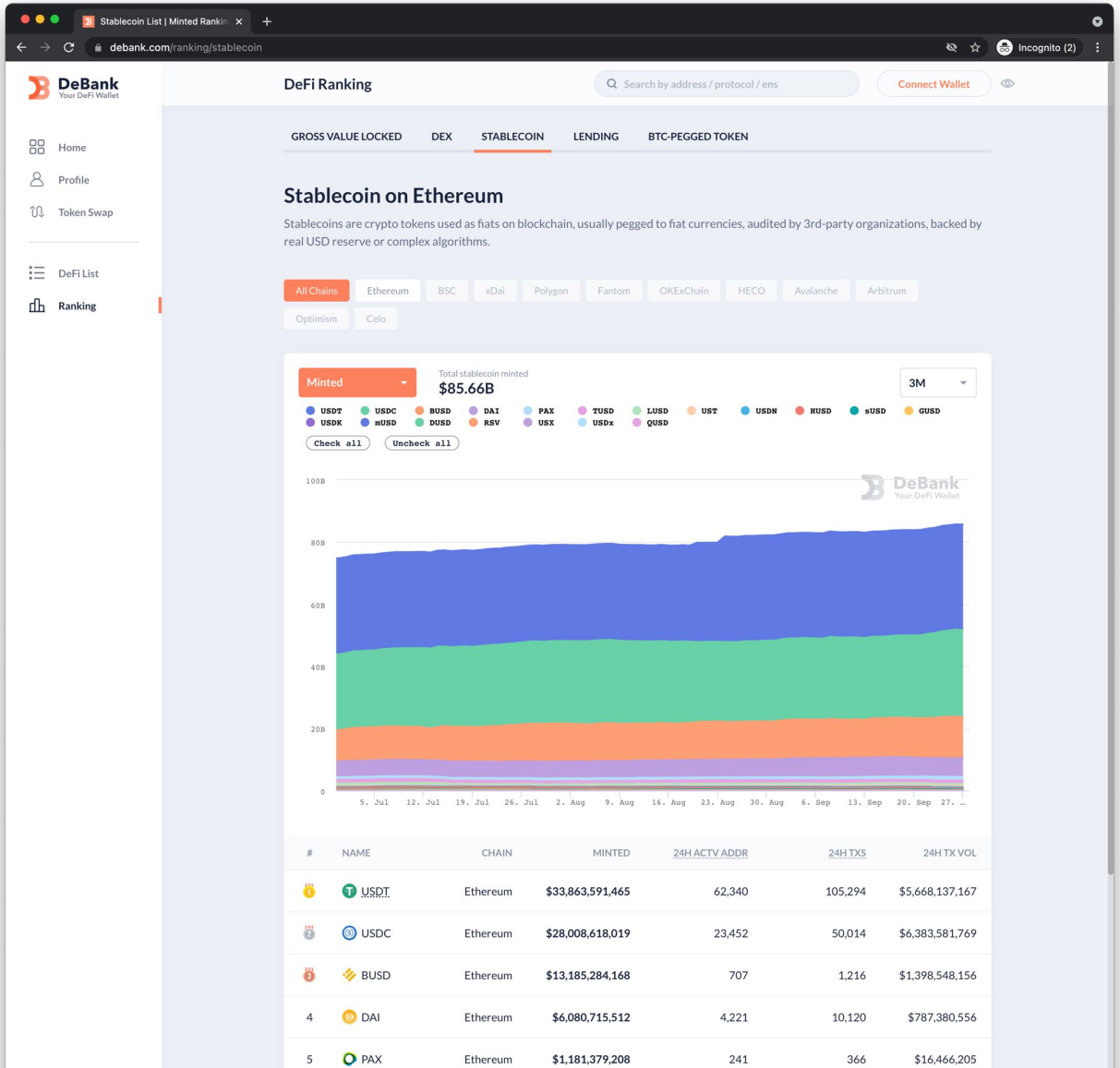
https://defi-learning.org

Economic Models

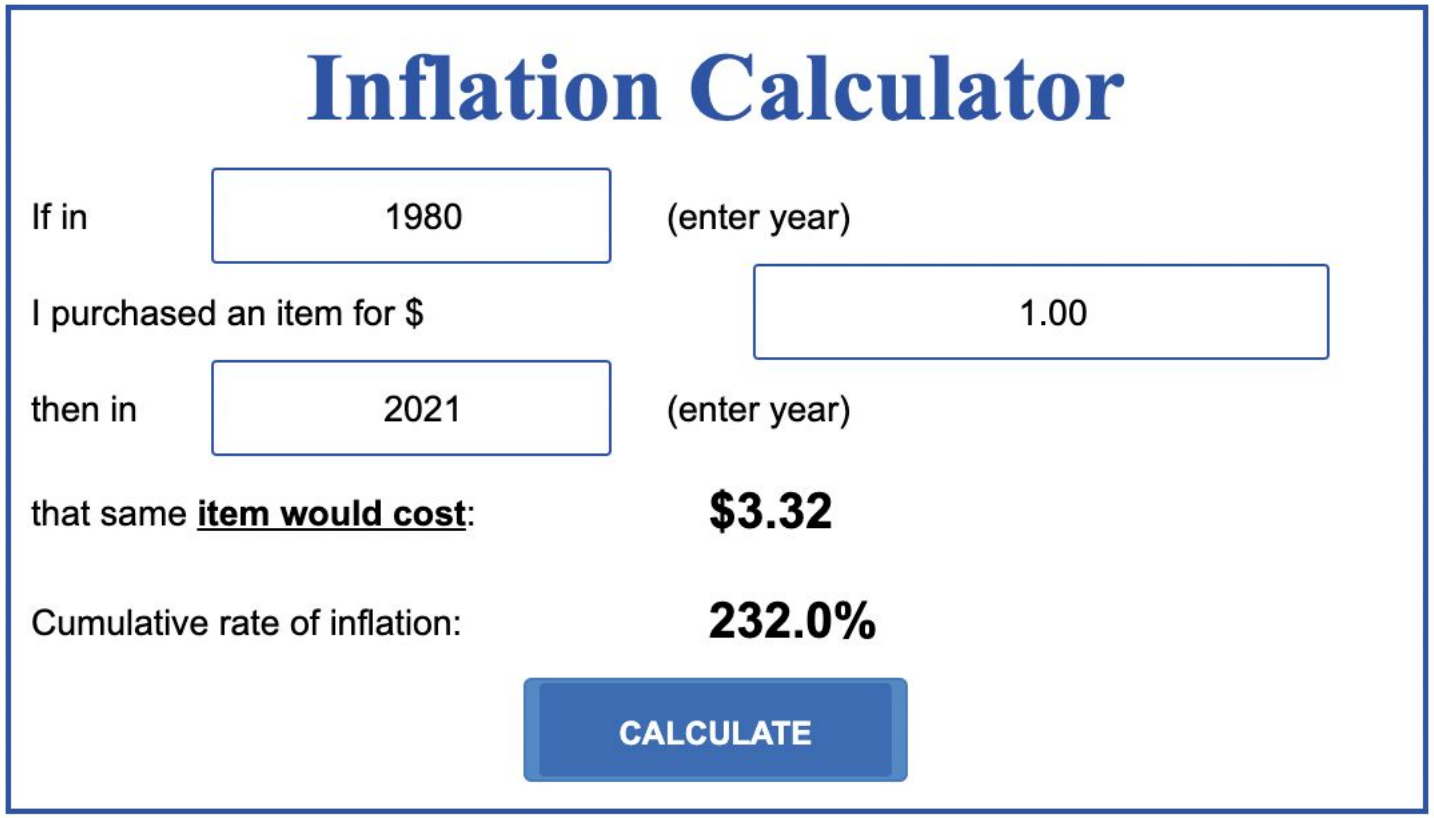
- You're given a model
  - Represents (often simplifies) an economic process
  - Contains variables
  - Logical/Quantitative relationships between variables

- Example: **Inflation**
  - Measuring inflation requires a *behavior model*
  - Differentiate relative vs. inflation price change

Stablecoins on Ethereum



Stable?



“Stablecoins” are not “stable”, but relatively more stable than other cryptocurrencies.

How do Stablecoins work?

MakerDAO

Synthetix

AMPL

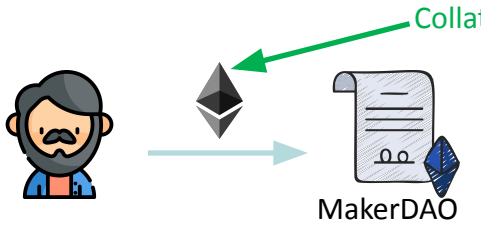
USDC

USDT

MakerDAO

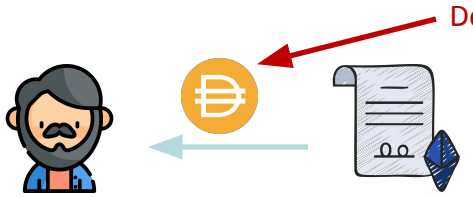
Stablecoin

Step 1



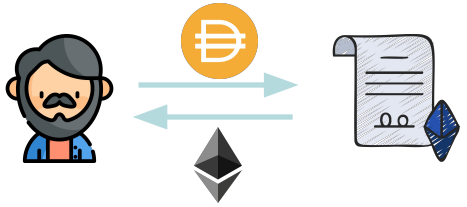
ETH deposited to open CDP
150% collateral mints 100% debt

Step 2



Draw DAI

Step 3

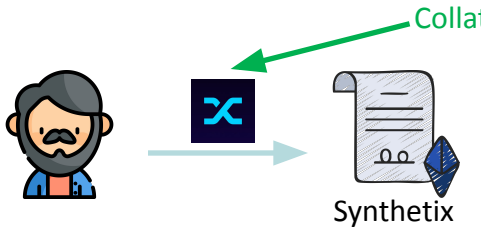


Pay back DAI to unlock ETH

Synthetix

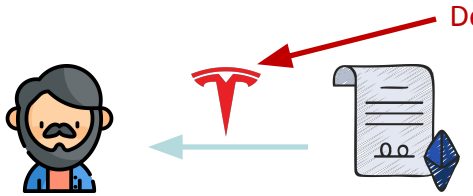
Mint a derivative asset

Step 1



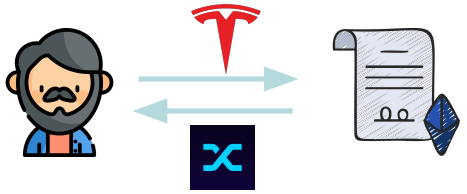
SNX deposited to mint derivative
600% collateral mints 100% derivative

Step 2



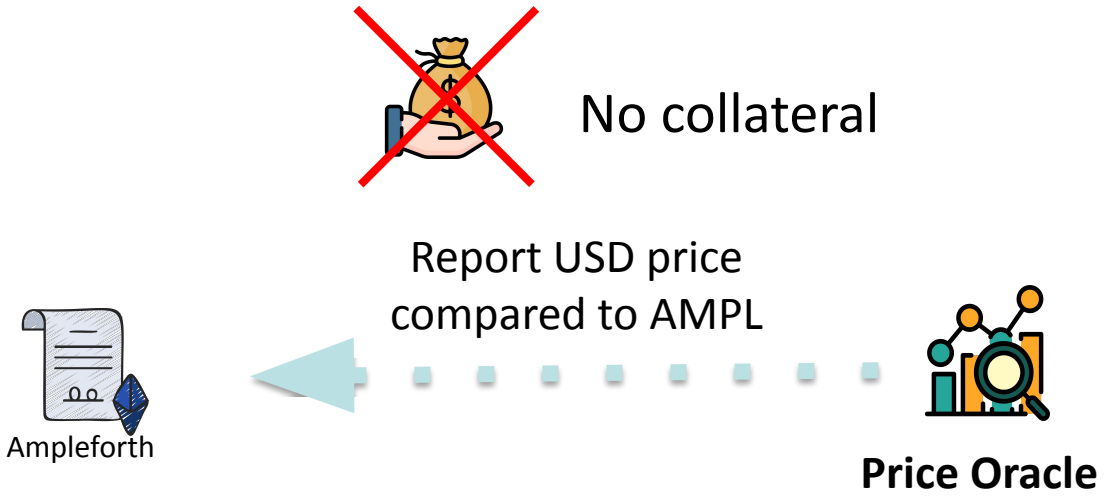
Draw derivative
(e.g. sTSLA share)

Step 3



Pay back sTSLA to unlock SNX

Ampleforth (AMPL)



- Three design states
  - Expansion: if 1 AMPL > 1 USD, add supply
  - Contraction: if 1 AMPL < 1 USD, remove supply
  - Equilibrium: no action taken

Systemic implications!

USDC and USDT

Company

Full or Fractional reserve

Mint coins

USDC/USDT

Add @ to blacklist

USDC/USDT

Destroy coins at @

USDT

USDT blacklisted over 400 accounts, +44M USDT destroyed.

USDC and USDT

```
1  function transfer(address _to, uint _value) public
       whenNotPaused {
2    require(!isBlackListed[msg.sender]);
3    if (deprecated) {
4      return UpgradedStandardToken(upgradedAddress).
           transferByLegacy(msg.sender, _to, _value);
5    } else {
6      return super.transfer(_to, _value);
7    }
8  }
9  function addBlackList (address _evilUser) public
       onlyOwner {
10   isBlackListed[_evilUser] = true;
11   AddedBlackList(_evilUser);
12 }
13 function destroyBlackFunds (address _blackListedUser)
       public onlyOwner {
14   require(isBlackListed[_blackListedUser]);
15   uint dirtyFunds = balanceOf(_blackListedUser);
16   balances[_blackListedUser] = 0;
17   _totalSupply -= dirtyFunds;
18   DestroyedBlackFunds(_blackListedUser, dirtyFunds);
19 }
```

DeFi Bank Run?

- CeFi Bank Run
  - Dangerous if fractional reserve
  - Most clients don't receive assets (or a minimum)
- DeFi Bank Run
  1. Event: USDT blacklists a pool, Stablecoin de-pegs
  2. Traders will exit pools
  3. Those who exit first receive the best prices
  4. Try to build blacklist detection bots